SOLAR FINANCIAL ROI & OUTPUT ASSESSMENT

ESTIMATED PERFORMANCE PROJECTION

NOTICE: AI-GENERATED CALCULATIONS & ESTIMATION BASIS

This assessment represents an automated financial and output projection derived directly from the provided system specification (Proposal & Quote QUO0000380). All figures—including annual yield, grid offset percentage, monthly cost savings, and payback timeline—are **theoretical estimates and operational assumptions**. Actual production and financial returns may vary based on site-specific weather patterns, future municipal tariff adjustments, and real household consumption behavior.

CLIENT	SITE ADDRESS	QUOTATION REF	TOTAL INVESTMENT
Pieter Bonthuys	340 Seaview Rd	QUO0000380	R149,500.00 (Incl. VAT)

15,685 kWh	~R5,000 / mo	2.3 – 2.5 Yrs	40.1% – 43.4%
EST. ANNUAL GENERATION	EST. YEAR 1 SAVINGS	SIMPLE PAYBACK PERIOD	YEAR 1 ROI RATE

QUOTED SYSTEM CONFIGURATION & CAPACITY

COMPONENT	SPECIFICATION DETAILS	CAPACITY / OUTPUT
Solar PV Array	16 × 580W Astronergy TOPCon Bifacial Panels	**9.28 kW** Total Peak Power
Hybrid Inverter	1 × Solis S6-EH1P12K03-NV-YD-L Single Phase	**12 kW** Power Rating
Battery Storage	1 × Kynee LiFePO4 Floor Standing Battery	**15.36 kWh** Storage Capacity
Mounting & BOS	Aluminium Solarstruts, Tile Hooks, AC/DC Protection	Full System Turnkey Kit

FINANCIAL YIELD & GRID OFFSET ANALYSIS

ASSESSMENT PARAMETER	CONSERVATIVE ESTIMATE (85% UTILISATION)	OPTIMISED ESTIMATE (92% UTILISATION)
Est. Annual Solar Yield	15,685 kWh / year	15,685 kWh / year
Est. Annual Grid Offset	~13,332 kWh / year	~14,430 kWh / year
Assumed Electricity Tariff	R4.50 / kWh (NMBM Residential)	R4.50 / kWh (NMBM Residential)
Est. Year 1 Financial Savings	**R59,994.00 / year**	**R64,935.00 / year**
Est. Average Monthly Offset	**~R4,999.50 / month**	**~R5,411.25 / month**
Simple Capital Payback Period	**~2.49 Years**	**~2.30 Years**
Escalation-Adjusted Payback (10% Annual Tariff Rise)	**~2.25 Years**	**~2.10 Years**

MODELING ASSUMPTIONS & CALCULATION METHODOLOGY

- **Generation Data Source:** Derived from the 15,685 kWh annual output calculation in Solar Proposal 340 Seaview Rd (modeled via System Advisor Model with 13.3% total system losses).
- **Grid Tariff Base:** Evaluated using a standard municipal benchmark rate of ~R4.50/kWh (incl. VAT) representative of current local NMBM residential tariffs.
- **Battery Offset Efficiency:** Assumes nighttime household base loads are substantially supplied by the 15.36 kWh battery bank, keeping overall daily self-consumption high.
- **Long-Term Value:** Hardware warranties include 15-year panel product (30-year performance) and 10-year battery warranties, ensuring net revenue generation for 20+ years post-payback.